

HumanFlow: Controllable Human Image Generation via Flow Matching

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Fig. 1. Results of HumanFlow for controllable human image generation. Human image generation is conditioned on the human-centric control conditions shown in the lower-left corner—keypoints, segmentation, Canny edges, surface normals, densePose, and depth—as well as the text prompts provided at the bottom.

We present HumanFlow, a unified flow-matching-based framework that enables high-fidelity and controllable full-body human image generation under diverse human-centric control conditions. Despite recent progress, controllable human image generation poses a fundamental challenge in balancing high visual fidelity with strict adherence to human-centric control conditions. HumanFlow formulates human image generation as a conditional flow-matching process with deterministic generation dynamics. To incorporate such human-centric control conditions into the pretrained model, we introduce a unified control framework with Control Encoder and Token-ControlNet. A Control Encoder maps diverse conditions into a unified latent

representation that is spatially aligned with the image latent space. Token-ControlNet is a lightweight control network architecturally aligned with the FLUX double-stream design. To address accurate structural control over human bodies, we further propose the Human Topology Consistency Loss (HTCL). HTCL regularizes conditional flow matching by constraining generated human configurations to a union of statistically grounded topology manifolds defined by normalized bone ratios and joint angles. To support large-scale training and systematic evaluation, we construct MiCoGen, a multi-condition human image dataset comprising over one million full-body human images with aligned text descriptions and rich human-centric control conditions. Extensive quantitative and qualitative evaluations on the MiCoGen dataset show that HumanFlow consistently achieves improved structural consistency than the existing diffusion-based and flow-matching-based methods, while maintaining high visual fidelity.

CCS Concepts: • Computing methodologies → Computer vision; Artificial intelligence.

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1 Introduction

Human image generation with fine-grained controllability has long been a central problem in computer graphics and computer vision.

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This is underpinning a wide range of applications such as digital human modeling [Sun et al. 2026; Xue et al. 2025; Yuanzhen Li 2024], virtual try-on systems [Guo et al. 2025a; Kim et al. 2024; Xu et al. 2025], and visual content creation [Choi et al. 2025a; Luo et al. 2024; Qian et al. 2025; Zheng et al. 2025, 2024]. Achieving precise control over human appearance, pose, and structure is essential for human image generation. However, existing methods [Jiang et al. 2022; Li et al. 2024a; Tan et al. 2025; Zhang et al. 2025b] struggle to simultaneously preserve high visual fidelity and enforce fine-grained structural constraints. To address these challenges, we propose HumanFlow that tightly couples semantic and structural conditioning within deterministic generation dynamics. HumanFlow enables fine-grained controllability without sacrificing visual fidelity.

Diffusion-based generative models [Ho et al. 2020; Rombach et al. 2022; Song et al. 2021] and text-to-image methods [Kawar et al. 2023; Labs et al. 2025; OpenAI 2024; Ruiz et al. 2023] have significantly improved visual realism and text-image semantic alignment, bringing generative models closer to practical deployment. But text-only supervision provides weak and indirect constraints on spatial structure, frequently leading to anatomically implausible poses or inconsistent body layouts. Moreover, existing conditional generation frameworks [Li et al. 2024b; Mou et al. 2024; Qin et al. 2023; Tan et al. 2025; Zhang et al. 2023] introduce external human-centric control conditions, such as keypoint or canny. However, such designs typically involve a trade-off between visual quality and control accuracy, making it difficult to simultaneously achieve high-quality and fine-grained controllable human image generation.

Flow-matching-based generative models [Albergo and Vanden-Eijnden 2022; Cao et al. 2025; Guo et al. 2025b; Lipman et al. 2023; Liu et al. 2025; Zhu et al. 2024] have recently emerged as a promising alternative to diffusion models. By directly supervising the velocity field along predefined interpolation paths, flow matching enables simplified and stable transport dynamics, often yielding training and inference pipelines that are more efficient than their diffusion-based counterparts [Gui et al. 2025; Liu et al. 2023; Xing et al. 2025]. However, existing flow-matching approaches predominantly focus on high-level semantic conditioning, such as text, and provide limited support for fine-grained, spatially structured control. In particular, the human-centric control conditions—e.g., depth, surface normals, or DensePose—to directly and consistently influence the learned flow dynamics remain largely unexplored.

In this work, we present HumanFlow, a flow-matching-based framework for controllable human image generation. It formulates image synthesis as a conditional flow-matching problem, learning a deterministic, time-dependent vector field under explicit human-centric control. The model generates images conditioned on a control signal and a text prompt. Compared to common diffusion-based control frameworks that rely on auxiliary modules for structural conditioning, HumanFlow integrates such conditions directly into the flow-matching formulation, yielding a more unified perspective on controllable human image generation.

To incorporate conditions into the pretrained model, we introduce Token-ControlNet. Token-ControlNet is a lightweight control mechanism architecturally aligned with the FLUX double-stream design. A dedicated Control Encoder first maps diverse control conditions into a unified latent representation that is spatially aligned

with the image latent space. The encoded control features are then injected into the generation process through token-level residual modulation. This design enables stable and temporally consistent conditioning while fully preserving the expressive capacity of the frozen backbone. To address accurate structural control over human bodies, we further propose the Human Topology Consistency Loss (HTCL). HTCL regularizes conditional flow matching by constraining generated human configurations to a union of statistically grounded topology manifolds defined by normalized bone ratios and joint angles. Instead of enforcing strict keypoint correspondence, HTCL constrains generated samples to remain in the manifold of anatomically plausible human configurations. As a result, HumanFlow exhibits improved stability under challenging scenarios such as extreme poses and partial occlusions.

To support large-scale training and systematic evaluation, we construct MiCoGen, a large-scale multi-condition human image dataset. MiCoGen contains approximately 1,019,124 full-body human images. Each image is paired with a text description and a set of control signals, including keypoints, Canny edges, depth maps, segmentation masks, DensePose, and surface normals. MiCoGen provides a unified benchmark for evaluating controllability, structural accuracy, and cross-modal consistency across diverse human-centric control conditions.

Our main contributions are summarized as follows:

- Propose HumanFlow, a flow-matching-based framework for controllable human image generation. HumanFlow formulates generation as conditional flow matching, where control signals guide the entire generation trajectory.
- Introduce a unified control mechanism with Token-ControlNet and HTCL. Token-ControlNet encodes diverse control signals into latent-aligned features and injects them into the generation process. HTCL leverages statistically grounded human body structure to guide conditional flow matching, promoting anatomically coherent and controllable human generation under diverse control signals.
- Construct MiCoGen, a large-scale multi-condition human image dataset with over 1M samples, each paired with text descriptions and spatially aligned control signals. The dataset provides a reliable benchmark for controllable human image generation.

We conducted extensive quantitative and qualitative evaluations on the proposed MiCoGen benchmark. Across a wide range of human-centric control conditions, HumanFlow consistently outperforms both diffusion-based and flow-matching-based methods in high visual quality and structural fidelity. In particular, it significantly reduces structural artifacts under challenging scenarios involving extreme poses and partial occlusions.

2 Related Work

2.1 Diffusion-Based Image Generation

Diffusion models have emerged as a dominant paradigm for image generation due to their strong expressiveness, training stability, and scalability [Dhariwal and Nichol 2021; Rombach et al. 2022; Song et al. 2021, 2020]. By formulating generation as an iterative denoising process that gradually transforms random noise into data

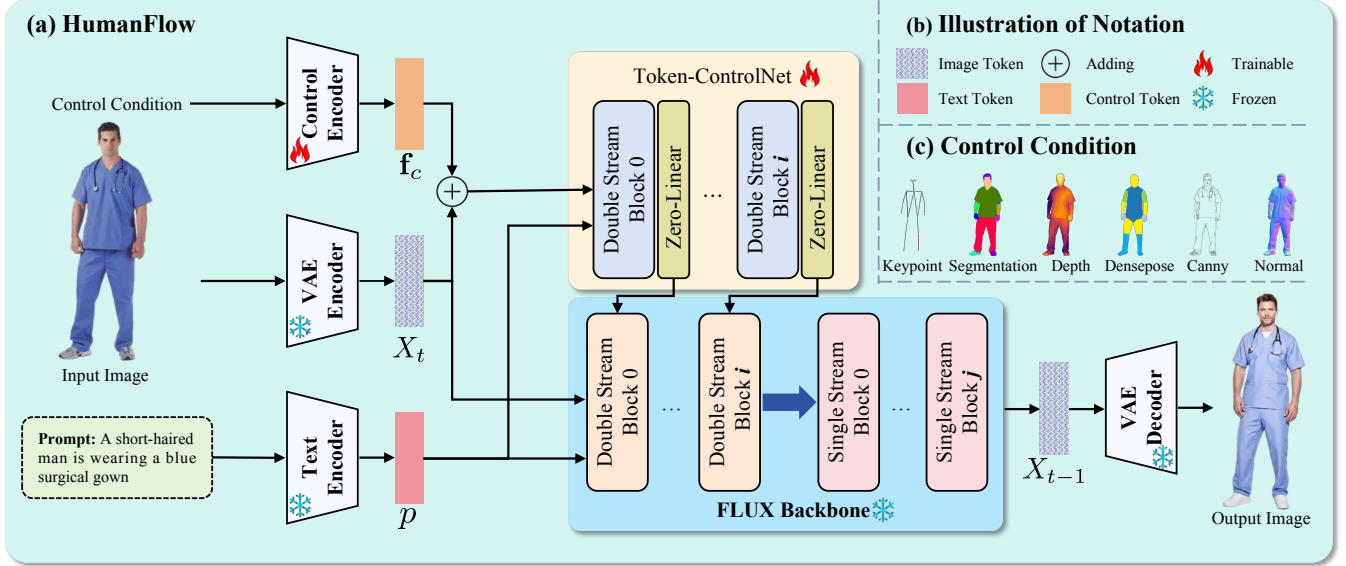


Fig. 2. (a) Overview of HumanFlow. Given a text prompt, an image and external control conditions, a frozen VAE encoder maps the image into latent image tokens X_t , while a frozen text encoder produces text tokens p . A trainable Control Encoder encodes the human-centric control conditions into control tokens f_c . At the token level, control tokens f_c is fused with image tokens X_t via element-wise addition and processed by the trainable Token-ControlNet. Token-ControlNet consists of a sequence of double-stream transformer blocks architecturally aligned with the frozen FLUX backbone, and generates layer-wise control residuals through zero-initialized linear projections. These residuals are injected into the corresponding double-stream blocks of the frozen backbone to modulate the flow-matching dynamics. Finally, a frozen VAE decoder maps the latent representation back to the output image. (b) Notation illustration, summarizing the symbols used in the overview. (c) Supported control conditions.

samples, diffusion-based approaches have achieved remarkable success in text-to-image synthesis [Bao et al. 2023; Kawar et al. 2023; Ruiz et al. 2023; Valevski et al. 2023] as well as conditional image generation [Huberman-Spiegelglas et al. 2024; Qian et al. 2025]. Diffusion models further improve visual fidelity and semantic alignment through extensive training data and classifier-free guidance strategies.

To enhance controllability, diffusion models have been extended to incorporate diverse conditional signals, including semantic layouts, edge maps, depth cues, keypoints representations, and other structural annotations. Representative methods [Bhat et al. 2024; Li et al. 2024b; Mou et al. 2024; Zhang et al. 2023] typically inject control information via auxiliary networks or modified denoising architectures, enabling the denoising process to follow external guidance. While effective, these approaches fundamentally rely on stochastic noise injection and long iterative denoising trajectories, which often introduce a trade-off between strict adherence to control conditions and overall visual quality. In particular, for articulated human generation, maintaining stable structural consistency remains challenging. Prior studies [Jin et al. 2025; Mou et al. 2024] show that guidance affects different aspects of generation across timesteps and tends to have diminishing influence in later iterations, making it non-trivial to maintain consistent controllability throughout the denoising trajectory.

2.2 Flow-Matching Based Image Generation

Flow-matching-based generative models [Lipman et al. 2023; Liu et al. 2023] have recently emerged as a deterministic alternative to diffusion models. Rather than simulating a stochastic denoising process, flow matching learns a time-dependent vector field that defines continuous generation dynamics between the source distribution and the target data distribution. The model is trained by directly supervising the velocity field along predefined interpolation paths, avoiding explicit likelihood estimation and handcrafted noise schedules. This formulation yields simplified generation trajectories and enables stable, continuous-time sampling.

Recent advances demonstrate that flow-matching-based models can achieve competitive performance in image [Liu et al. 2025; Zhu et al. 2024] and video generation [Ki et al. 2025; Wang et al. 2024a; Zhang et al. 2025a]. Extensions to conditional and cross-modal generation further show that high-level semantic alignment, such as text-to-image synthesis, can be effectively modeled within a unified flow-matching framework [Liu et al. 2025]. However, existing methods predominantly focus on semantic conditioning and offer limited support for spatially structured control signals. In particular, principled mechanisms that allow conditions such as pose, depth, or other geometric cues to directly influence the learned flow dynamics remain largely unexplored. As a result, the applicability of current flow-matching-based models to fine-grained, structurally controllable generation tasks is still limited.

2.3 Human Image Generation

Human image generation has been extensively studied due to its importance in digital humans [Li et al. 2024a; Sun et al. 2026; Xue et al. 2025], virtual try-on [Han et al. 2018; Huang et al. 2025; Kim et al. 2024; Wang et al. 2025; Xu et al. 2025], and personalized content creation [Huang et al. 2024; Qian et al. 2025]. Early works [Fu et al. 2022; Jiang et al. 2022; Ju et al. 2023; Li et al. 2024a; Wang et al. 2024b] primarily focus on generating human images from text descriptions, where text prompts are used to specify clothing attributes, appearance styles, and global semantics. While such approaches can produce visually plausible results, text-based supervision alone often provides only coarse-grained control and struggles to capture fine-grained structural variations of the human body.

To improve controllability, a line of research [Jiang et al. 2022; Ju et al. 2023; Li et al. 2024a] introduces additional structural conditions, such as keypoints representations or body part annotations, to guide human image generation. Test-time optimization or fine-tuning strategies further enhance customization quality by adapting models to specific identities or appearances. Although these methods can achieve satisfactory results for individual aspects of human appearance, they often incur significant computational overhead and lack scalability when multiple parts or conditions must be controlled simultaneously.

More recently, reference-based human image generation methods [Choi et al. 2025b; Huang et al. 2024; Ji et al. 2025; Qian et al. 2025; Zhou et al. 2025] have demonstrated promising results by encoding identity-related information—such as facial appearance—from one or more reference images into compact representations. These approaches enable rapid personalization without per-instance training and can be extended with structural guidance to support keypoints control or identity preservation. Among them, Leffa [Zhou et al. 2025] explores reference-driven human image generation by learning flow fields within attention modules in diffusion models, where reference images provide explicit guidance for appearance and pose. In contrast, HumanFlow formulates human image generation as a conditional flow-matching process, guided by a control signal and a text prompt, without relying on reference images. However, they typically rely on limited visual cues from a small number of reference images, which can lead to an ambiguous or incomplete representation of human subjects. As a result, maintaining consistent details across multiple body parts and satisfying diverse control conditions simultaneously remains a challenge.

Overall, diffusion-based approaches are difficult for high-fidelity human images. Flow-matching-based methods primarily emphasize high-level semantic alignment and provide limited support for human-centric control conditions. These limitations lead to high-fidelity and controllable full-body human image generation under diverse human-centric control conditions. This motivates the need for a unified human image generative framework that can support diverse human-centric control conditions.

3 Method

We propose HumanFlow, a flow-matching-based framework for controllable human image generation. The overall pipeline is illustrated in Figure 2. HumanFlow formulates generation as a conditional

continuous-time transport process, in which human-centric control conditions explicitly influence the learned flow dynamics. To this end, we introduce a token-level control mechanism that injects structured conditions into a frozen flow backbone via lightweight residual pathways, together with a topology-aware regularization that enforces anatomically consistent human structures. To support learning and evaluation under diverse control conditions, we construct MiCoGen, a large-scale multi-condition human image dataset with over one million full-body samples, each paired with text descriptions and spatially aligned human-centric annotations.

3.1 Preliminaries

We consider a generative modeling framework that defines a continuous transformation from a simple source distribution to a target data distribution via a deterministic ordinary differential equation (ODE). In this formulation, samples are transported along a time-dependent vector field over a continuous time interval $t \in [0, 1]$.

Let $\mathcal{X} \subset \mathbb{R}^d$ denote the image space. We denote by $\mathbf{x}_0 \sim p_0(\mathbf{x})$ a sample drawn from a simple source distribution (e.g., a standard Gaussian), and by $\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})$ a sample drawn from the data distribution.

Flow Matching aims to learn a deterministic, time-dependent vector field:

$$\mathbf{v}_\theta : \mathcal{X} \times [0, 1] \rightarrow \mathbb{R}^d, \quad (1)$$

which is parameterized by θ , such that the induced dynamics transport samples from p_0 to p_{data} by solving the ODE:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_\theta(\mathbf{x}(t), t), \quad \mathbf{x}(0) = \mathbf{x}_0. \quad (2)$$

Rather than modeling generation as a stochastic denoising process, flow matching directly supervises the vector field along predefined interpolation paths between samples from the source and data distributions. A commonly adopted choice is linear interpolation:

$$\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1, \quad t \sim \mathcal{U}(0, 1), \quad (3)$$

which induces a constant target velocity:

$$\mathbf{u}(\mathbf{x}_t, t) = \frac{d\mathbf{x}_t}{dt} = \mathbf{x}_1 - \mathbf{x}_0. \quad (4)$$

The flow-matching objective minimizes the expected squared error between the predicted velocity and the target velocity along the interpolation path:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{x}_1 \sim p_{\text{data}}, t \sim \mathcal{U}(0, 1)} [\|\mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{u}(\mathbf{x}_t, t)\|_2^2]. \quad (5)$$

Minimizing $\mathcal{L}_{\text{FM}}$ encourages the learned vector field to match the prescribed transport dynamics in expectation, yielding an ODE whose solution maps samples from the source distribution toward the data distribution at $t = 1$.

3.2 Controllable Flow Matching

The standard flow-matching formulation models unconditional transport dynamics. For controllable human image generation, however, the generation process must explicitly account for structured, human-centric control conditions throughout the continuous-time evolution.

Let $\mathbf{c} \sim p_{\mathbf{c}}(\mathbf{c} \mid \mathbf{x}_1)$ denote a control signal associated with the target image $\mathbf{x}_1$. The control variable $\mathbf{c}$ may represent keypoints,

edge maps, depth cues, densepose annotations, or other structural representations related to human topology.

HumanFlow extends flow matching by learning a conditional vector field:

$$\mathbf{v}_\theta(\mathbf{x}, t \mid \mathbf{c}), \quad (6)$$

which defines control-dependent transport dynamics governed by the ODE:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_\theta(\mathbf{x}(t), t \mid \mathbf{c}), \quad \mathbf{x}(0) = \mathbf{x}_0. \quad (7)$$

Training follows the same interpolation strategy as in Eq. (3), while the control signal is incorporated directly into the definition of the vector field. The conditional flow-matching objective is therefore given by:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{\mathbf{x}_0, \mathbf{x}_1, \mathbf{c}, t} [\|\mathbf{v}_\theta(\mathbf{x}_t, t \mid \mathbf{c}) - \mathbf{u}(\mathbf{x}_t, t \mid \mathbf{x}_0, \mathbf{x}_1)\|_2^2]. \quad (8)$$

By conditioning the vector field, HumanFlow allows control signals to guide the entire continuous transport process. This is crucial for human image generation, where global structural consistency must be maintained. Unlike diffusion-based approaches, where guidance may vary across timesteps, flow matching models generate a continuous trajectory, providing a more direct and stable mechanism for enforcing structural coherence and controllability.

3.3 Token-ControlNet

Directly modifying or fine-tuning a pretrained flow backbone often disrupts the learned vector field, leading to unstable trajectories, which is particularly problematic for human generation. To address this, Token-ControlNet introduces a lightweight, parallel control pathway that injects multi-modal structural conditions at the token level via layer-wise residuals. By keeping the FLUX backbone frozen, this design preserves its expressive capacity and stability, while enabling precise and temporally consistent controllability over human-centric conditions.

Control Encoder. Controllable human image generation requires structural conditions to influence the generative process in a spatially aligned and stable manner. To this end, HumanFlow employs a dedicated Control Encoder E_{ctrl} that maps diverse control conditions into feature representations aligned with the latent space of the flow-matching backbone.

Let $\mathbf{c} \in \mathbb{R}^{B \times C \times H \times W}$ denote an input control condition, where C is the modality-specific channel dimension. The Control Encoder produces a control feature map:

$$\mathbf{f}_c \in \mathbb{R}^{B \times C' \times H' \times W'}. \quad (9)$$

The encoder follows a hierarchical convolutional architecture. It starts with a 3×3 convolution and a residual block [He et al. 2016] to extract low-level structural cues. Subsequent stages apply stride-2 3×3 convolutions followed by residual blocks, progressively enlarging the receptive field while maintaining spatial coherence. To accommodate diverse control conditions, channel attention is applied at early layers for modality-aware reweighting, local spatial attention is introduced at intermediate stages to emphasize fine-grained structures, and global attention is employed at deeper layers to capture long-range dependencies and holistic human topology.

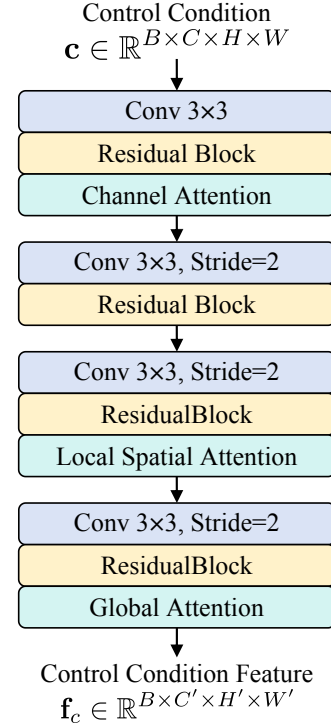


Fig. 3. Overview of the Control Encoder. The Control Encoder maps the multi-channel control condition into spatially aligned latent features using stacked 3×3 convolutions and residual blocks [He et al. 2016] with progressive stride-2 downsampling. Channel attention is applied in early layers for modality-aware reweighting, followed by local spatial attention at intermediate stages to emphasize fine-grained structures, and global attention in deeper layers to capture long-range dependencies and holistic human topology cues. The encoder outputs a control feature map aligned with the VAE latent resolution at $H' \times W'$.

The resulting control features play two complementary roles. First, they are aligned and fused with latent image features to form vision tokens for joint modeling with text tokens. Second, they are injected into the conditional flow dynamics through Token-ControlNet, ensuring that control conditions influence the generation trajectory throughout continuous time.

Token-Level Control Injection. Token-ControlNet is architecturally aligned with the FLUX backbone. Specifically, the control pathway replicates the structure of the FLUX double-stream blocks from Double Stream Block 0 to Double Stream Block i , where each control block mirrors the corresponding backbone block in terms of token layout and attention structure. This one-to-one alignment ensures that control features are injected at semantically and temporally consistent stages of the flow-matching dynamics. By restricting control injection to the early double-stream blocks, Token-ControlNet imposes strong structural guidance at the initial stages of generation, where global layout and human topology are established. Meanwhile, it preserves the high-level semantic modeling capacity of the deeper backbone layers.

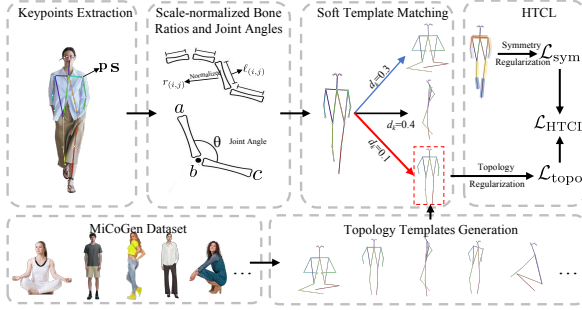


Fig. 4. Overview of the Human Topology Consistency Loss (HTCL). Given a generated human image, a frozen keypoints estimator extracts 2D keypoints. Human topology is encoded using scale-normalized bone-length ratios and joint angles defined on the skeletal graph. These features are compared against a data-driven multi-template topology prior, with relaxed symmetry regularization, to suppress implausible body configurations.

Token-ControlNet injects control information into the flow-based generation process via token-level residual modulation. Given a control condition $\mathbf{c}$, the Control Encoder produces a spatially aligned feature:

$$\mathbf{f}_c = E_{\text{ctrl}}(\mathbf{c}) \in \mathbb{R}^{B \times C' \times H' \times W'}, \quad (10)$$

where (H', W') matches the spatial resolution of the image latent representation. The control feature is reshaped into a sequence of control tokens $\mathbf{t}_c \in \mathbb{R}^{B \times N \times D}$ and fused with image tokens $\mathbf{t}_x$ via element-wise addition:

$$\tilde{\mathbf{t}}_x = \mathbf{t}_x + \mathbf{t}_c. \quad (11)$$

The control-augmented image tokens $\tilde{\mathbf{t}}_x$ are jointly processed with text tokens $\mathbf{p}$ and time-step embeddings by a small number of double-stream transformer blocks aligned with the frozen FLUX backbone. Specifically, Token-ControlNet injects control signals into the double-stream blocks from index $l = 0$ to $l = i$.

At each controlled block l , Token-ControlNet produces a control residual through a linear projection initialized to zero:

$$\Delta \mathbf{t}_x^{(l)} = \mathbf{h}^{(l)} \mathbf{W}_l^\top + \mathbf{b}_l, \quad \mathbf{W}_l^{(0)} = \mathbf{0}, \mathbf{b}_l^{(0)} = \mathbf{0}, \quad (12)$$

where $\mathbf{h}^{(l)} \in \mathbb{R}^{B \times N \times D}$ denotes the intermediate control features at block l , and $\mathbf{W}_l^{(0)} = \mathbf{0}$ and $\mathbf{b}_l^{(0)} = \mathbf{0}$ denote that the linear projection parameters are initialized to zero at the start of training.

The control residual is injected into the frozen backbone via a residual connection:

$$\mathbf{t}_x^{(l+1)} = \mathbf{t}_x^{(l)} + \gamma \Delta \mathbf{t}_x^{(l)}, \quad l = 0, \dots, i, \quad (13)$$

where γ controls the strength of the control signal.

The zero-initialization strategy ensures that all control residuals are identically zero at the beginning of training, leaving the pretrained FLUX generation dynamics unchanged. As training progresses, the control pathway gradually learns meaningful, structure-aware transformations, enabling precise spatial control while maintaining the stability, interpretability, and deterministic dynamics of the underlying flow-matching model.

3.4 Human Topology Consistency Loss

To encourage structurally coherent human configurations and suppress implausible body layouts during generation, we introduce the Human Topology Consistency Loss (HTCL), as shown in Figure 4. HTCL acts as a geometry-aware regularization term that constrains the generative process to remain consistent with statistically valid human body topology. Importantly, HTCL does not use absolute keypoint locations. This design is particularly effective under extreme poses, partial occlusions, or ambiguous control signals, where direct keypoint supervision becomes unreliable. HTCL models scale-normalized bone-ratio and joint-angle, which describe the underlying skeletal topology. This representation keeps anatomical consistency, making human structures more robust.

Given a generated image $\mathbf{x}$, we extract 2D human keypoints $\mathbf{p} \in \mathbb{R}^{J \times 2}$ together with confidence scores $\mathbf{s} \in [0, 1]^J$ using a frozen keypoints estimator with COCO-format keypoints ($J = 17$). The human body is represented by a sparse skeleton graph $\mathcal{E}$, where each edge corresponds to a body bone, and a set of angle triplets $\mathcal{T}$ that encode joint-level directional relations.

Topology Feature Representation. HTCL characterizes human topology using scale-normalized bone ratios and joint angles. For each bone $(i, j) \in \mathcal{E}$, the bone length is defined as $\ell_{(i,j)} = \|\mathbf{p}_j - \mathbf{p}_i\|_2$. To obtain a scale-invariant representation, we estimate a robust torso scale using a predefined set of torso bones $\mathcal{E}_{\text{torso}}$:

$$s_{\text{torso}} = \max \left(\frac{1}{|\mathcal{E}_{\text{torso}}|} \sum_{(u,v) \in \mathcal{E}_{\text{torso}}} \ell_{(u,v)}, s_{\min} \right), \quad (14)$$

where $s_{\min}$ prevents numerical instability under extreme foreshortening. We use a separate index (u, v) to denote bones in the torso subset $\mathcal{E}_{\text{torso}}$ for scale estimation, to avoid ambiguity with the generic bone index $(i, j) \in \mathcal{E}$ used in subsequent topology computations. The normalized bone ratio is then computed as:

$$r_{(i,j)} = \frac{\ell_{(i,j)}}{s_{\text{torso}} + \epsilon}, \quad (i, j) \in \mathcal{E}. \quad (15)$$

For each joint triplet $(a, b, c) \in \mathcal{T}$, we compute the joint angle at vertex b :

$$\theta_{(a,b,c)} = \arccos \left(\frac{\langle \mathbf{p}_a - \mathbf{p}_b, \mathbf{p}_c - \mathbf{p}_b \rangle}{\|\mathbf{p}_a - \mathbf{p}_b\|_2 \|\mathbf{p}_c - \mathbf{p}_b\|_2 + \epsilon} \right) \in [0, \pi]. \quad (16)$$

To reduce the influence of unreliable detections, confidence-based masks are applied:

$$\mathbf{1}_i = \begin{cases} 1, & s_i > \delta, \\ 0, & \text{otherwise.} \end{cases} \quad (17)$$

$$m_{(i,j)}^r = \mathbf{1}_i \mathbf{1}_j, \quad (18)$$

$$m_{(a,b,c)}^\theta = \mathbf{1}_a \mathbf{1}_b \mathbf{1}_c. \quad (19)$$

Here, δ denotes a confidence threshold used to filter out unreliable keypoint detections predicted by the pose estimator.

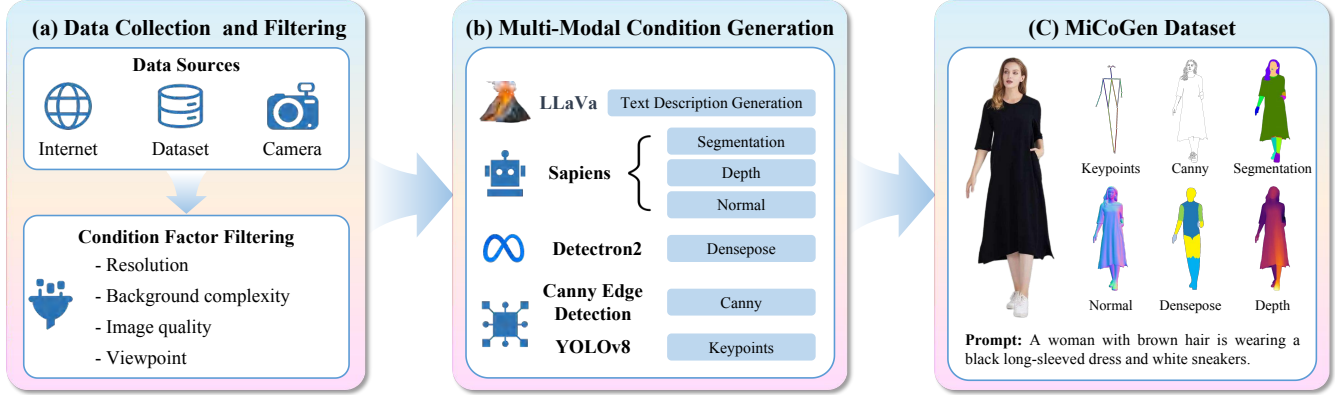


Fig. 5. MiCoGen Dataset Construction Pipeline. (a) Data Collection and Filtering: Raw images are collected from the internet, existing datasets, and camera captures, and are subsequently filtered according to resolution, background complexity, image quality, and viewpoint. (b) Multi-Modal Condition Generation: For the filtered images, multi-modal conditions are automatically generated, including text descriptions (via LLaVA), human-centric control conditions such as segmentation, depth, densepose, and normals (via Sapiens [Khrodar et al. 2024] and Detectron2 [Wu et al. 2019]), as well as Canny edge and keypoints. (c) MiCoGen Dataset: The processed images and their aligned multi-modal annotations are assembled into the MiCoGen dataset.

Multi-Template Topology Prior. Human body configurations exhibit substantial variability, which a single canonical template cannot adequately model. HTCL therefore adopts a multi-template formulation. HTCL will project generated samples onto a union of coarse human topology manifolds, instead of enforcing conformity to a single canonical template. Skeleton features extracted from training data are clustered into K keypoints families, yielding a set of topology templates $\{(\bar{\mathbf{r}}_k, \bar{\theta}_k)\}_{k=1}^K$. Here, $\bar{\mathbf{r}}_k$ and $\bar{\theta}_k$ denote the mean normalized bone-ratio vector and mean joint-angle vector of the k -th topology template, respectively, estimated from the training data. Each topology template captures a coarse yet statistically stable human pose family, encoding typical limb proportions and joint configurations while remaining invariant to global scale, viewpoint, and articulation-specific noise. The number of templates K is determined by balancing structural diversity and statistical reliability, rather than matching predefined pose categories. Specifically, bone-ratio and joint-angle ranges characterize the intra-template structural variation, while topology weights reflect the proportion of samples assigned to each template. As shown in Table 1, a small K (e.g., $K = 3$) leads to overly broad templates that merge structurally diverse poses, whereas a large K (e.g., $K = 7$) yields insufficient samples per template, reducing statistical stability. Therefore, $K = 5$ achieves a favorable trade-off, maintaining moderate structural variation with well-supported templates. Accordingly, we set $K = 5$ in HumanFlow.

Table 1. Structural statistics under different numbers of topology templates.

K	Topology weight range	Bone-ratio range	Joint-angle range
3	0.24–0.42	0.574–3.004	20.4°–123.7°
5	0.16–0.28	0.410–2.549	16.1°–123.4°
7	0.12–0.22	0.403–2.014	12.2°–119.6°

For a generated sample with features $(\mathbf{r}, \theta)$, its distance to template k is measured using masked, weighted errors:

$$d_k = \lambda_r \frac{\sum_{e \in \mathcal{E}} m_e^r w_e^r (r_e - \bar{r}_{k,e})^2}{\sum_{e \in \mathcal{E}} m_e^r + \varepsilon} + \lambda_\theta \frac{\sum_{t \in \mathcal{T}} m_t^\theta w_t^\theta (\theta_t - \bar{\theta}_{k,t})^2}{\sum_{t \in \mathcal{T}} m_t^\theta + \varepsilon}, \quad (20)$$

which m_e^r and m_t^θ are confidence-based binary masks indicating whether the involved keypoints are reliable, and they exclude low-confidence bones or joints from the loss computation.

To avoid a non-differentiable hard assignment, we aggregate template distances using a soft-min operator:

$$\mathcal{L}_{\text{topo}} = -\tau \log \sum_{k=1}^K \exp\left(-\frac{d_k}{\tau}\right), \quad (21)$$

where τ controls the smoothness of the approximation.

Symmetry Regularization. To discourage extreme left–right asymmetry while remaining robust to viewpoint and perspective effects, we introduce a relaxed symmetry constraint. Given a set of symmetric bone pairs $\mathcal{P}$, the symmetry loss is defined as:

$$\mathcal{L}_{\text{sym}} = \frac{1}{|\mathcal{P}|} \sum_{(e_L, e_R) \in \mathcal{P}} \max\left(0, \frac{\max(r_{e_L}, r_{e_R})}{\min(r_{e_L}, r_{e_R}) + \varepsilon} - \alpha\right) \cdot m_{e_L}^r m_{e_R}^r, \quad (22)$$

where α specifies an allowable asymmetry margin.

Overall HTCL Objective. The final Human Topology Consistency Loss is given by:

$$\mathcal{L}_{\text{HTCL}} = \mathcal{L}_{\text{topo}} + \lambda_{\text{sym}} \mathcal{L}_{\text{sym}}, \quad (23)$$

which is jointly optimized with the conditional flow-matching objective to promote topology-consistent and controllable human image generation. HTCL is applied only during training. Unless otherwise specified, we set the keypoint confidence threshold to $\delta = 0.3$, use $K = 5$ topology templates, apply a soft-min temperature $\tau = 0.1$, weight the bone and angle terms by $\lambda_r = 1.0$ and $\lambda_\theta = 0.3$ respectively, and set the symmetry regularization weight to $\lambda_{\text{sym}} = 0.2$.

Table 2. Comparison of representative human image datasets and their available annotation conditions. MiCoGen uniquely provides comprehensive human-centric annotations, including text descriptions, keypoints, segmentation, depth, densepose, surface normals, and Canny edges, enabling fine-grained and fully controllable human image generation.

Dataset	Quantity	Annotation						
		Text	Keypoints	Segmentation	Depth	DensePose	Normal	Canny
DF-MM [Jiang et al. 2022]	44K	✓	✓	✓	×	✓	×	×
SHHQ 1.0 [Fu et al. 2022]	40K	×	✓	✓	×	×	×	×
LAION-Human [Ju et al. 2023]	1M	✓	×	×	×	×	×	×
CosmicMan-HQ 1.0 [Li et al. 2024a]	6M	✓	×	✓	×	×	×	×
MiCoGen	1M	✓	✓	✓	✓	✓	✓	✓

3.5 MiCoGen Dataset

To support large-scale training and systematic evaluation of controllable human image generation, we construct MiCoGen, a multi-condition human image dataset specifically designed for conditional flow-matching frameworks. MiCoGen contains 1,019,124 full-body human images, each paired with an automatically generated text description and a diverse set of spatially aligned human-centric control conditions, including keypoints, segmentation masks, densepose, depth maps, surface normals, and Canny edges.

Data Collection and Filtering. Images are collected from a combination of publicly available internet resources, existing datasets (including IGPair [Huang et al. 2024], CosmicManHQ-1.0 [Li et al. 2024a], LAION-Human [Ju et al. 2023], and DeepFashion-MultiModal [Jiang et al. 2022]), as well as in-house captured data. To ensure data quality and suitability for controllable generation, all images undergo a multi-stage filtering pipeline. Specifically, we retain only high-resolution, full-body human images with clear visual appearance and minimal occlusion. Images with severe truncation, extreme viewpoints, or excessive background clutter are discarded. All retained images are normalized to a unified spatial resolution to facilitate the generation of spatially aligned control signals.

Text Description Generation. For each retained image, a descriptive text prompt is automatically generated using LLaVA. The generated captions focus on high-level semantic attributes, such as clothing type, color, style, and overall appearance. These text descriptions serve as semantic conditions during training and enable text-guided human image generation within the HumanFlow framework.

Human-Centric Control Conditions Generation. To enable fine-grained structural controllability, we generate multiple human-centric control conditions for each image. We employ Sapiens [Khironkar et al. 2024], a family of models specialized for fundamental human-centric vision tasks, to predict body-part segmentation maps, depth maps, and surface normal maps. Additionally, we utilize Detectron2 [Wu et al. 2019] to generate densepose annotations. Human keypoints are extracted using YoloV8 [Talaat and ZainEldin 2023]. In addition, Canny edge maps are computed using the standard Canny edge detector to capture contour-level structural information. All predicted control conditions are spatially aligned with their corresponding images and normalized to a unified coordinate system, ensuring consistent and reliable conditioning across conditions.

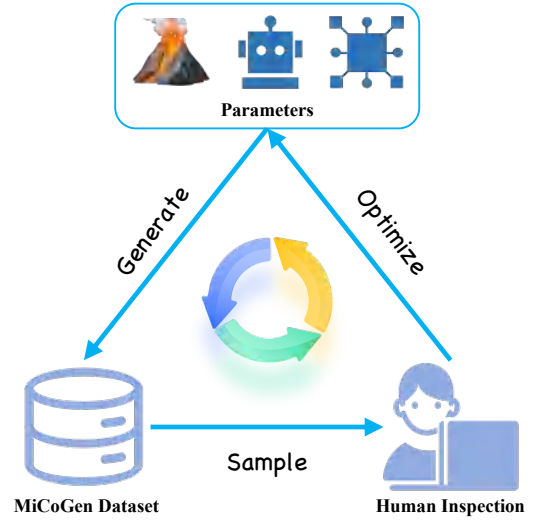


Fig. 6. Overview of the iterative pipeline for MiCoGen.

Dataset iteration. We ensure annotation quality through a multi-stage pipeline with iterative human verification, as illustrated in Fig. 6. We randomly sample 10,000 pairs for manual inspection. Then, we manually identify common errors, including misaligned keypoints, missing body parts, and inconsistent geometric signals. The next step is to optimize the data generation process. For text descriptions, we improve LLaVA prompts and adjust generation parameters, including temperature, num_predict, and repeat_penalty. For control signals, we optimize modality-specific settings, including keypoint confidence thresholds, Canny thresholds, and model weights for Sapiens [Khironkar et al. 2024]. This process is iterative. After each refinement, a new subset is sampled for verification. The pipeline is finalized once the annotation validity exceeds 98%.

Comparison. As shown in the table 2, MiCoGen is distinguished not only by its scale but, more importantly, by the completeness and alignment of its human-centric control conditions. Prior datasets typically focus on a limited subset of conditions—such as keypoints or segmentation—or provide only weak semantic supervision through text, which constrains their applicability to fine-grained controllable generation. In contrast, MiCoGen simultaneously provides dense

geometric cues (e.g., depth, normals, densepose), sparse structural representations (e.g., keypoints and edges), and high-level semantic descriptions for each image, all spatially aligned in a unified coordinate system. This comprehensive multi-modal design enables systematic training and evaluation of conditional generation models under diverse control signals.

3.6 Unified Training Objective

HumanFlow is trained by jointly optimizing conditional flow matching and a human-centric topology regularization. Given a target image $\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})$ and its associated control condition $\mathbf{c} \sim p_{\mathcal{C}}(\mathbf{c} | \mathbf{x}_1)$, we sample a source latent $\mathbf{x}_0 \sim p_0(\mathbf{x})$ and a time variable $t \sim \mathcal{U}(0, 1)$. An intermediate latent state $\mathbf{x}_t$ is constructed via linear interpolation following Eq. (3).

The conditional vector field $\mathbf{v}_\theta(\mathbf{x}, t | \mathbf{c})$ is learned by minimizing the conditional flow-matching objective Eq. (8). This objective encourages the learned vector field to realize control-dependent transport dynamics that map the source distribution to the data distribution over continuous time.

To further promote anatomically coherent and structurally plausible human configurations under diverse control signals, we incorporate the proposed Human Topology Consistency Loss as a geometry-aware regularization term. HTCL is computed by Eq. (23).

The overall training objective is given by a weighted combination:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{CFM}}(\theta) + \lambda_{\text{HTCL}} \mathcal{L}_{\text{HTCL}}(\theta), \quad (24)$$

where λ_{HTCL} controls the strength of topology regularization. λ_{HTCL} is set to 0.3.

During optimization, the pretrained FLUX backbone and the VAE and text encoders are kept fixed. Only the Control Encoder and Token-ControlNet parameters are updated under $\mathcal{L}_{\text{total}}$, ensuring that structural controllability is learned through lightweight residual pathways while preserving the stability and expressive capacity of the underlying flow-matching model.

4 Experiments

4.1 Experimental Setup

Implementation Details. HumanFlow is built upon the FLUX.1-dev [Labs 2024] backbone with a double-stream transformer architecture, using 64 input channels, hidden size 3072, 24 attention heads, and 19 double-stream blocks. Text conditioning is provided by CLIP [Radford et al. 2021] and T5 [Raffel et al. 2020]. The VAE adopts a standard encoder-decoder with a latent dimension 16. We employ Token-ControlNet, which shares the backbone feature dimensions. A Control Encoder maps each condition into spatial features aligned with the latent space.

We train on MiCoGen. Images and control signals are resized to 512×512 without cropping to preserve the full-body structure and then normalized. All control signals are spatially aligned with the corresponding images. Canny images are converted to three-channel inputs. Data loading follows standard random shuffling and batching. HTCL operates on COCO 17-point keypoints using scale-normalized bone ratios and joint angles. We set $K = 5$ topology templates, obtained via K-means clustering over normalized bone-angle features. HumanFlow is implemented in PyTorch and trained

on a single NVIDIA RTX PRO 6000 GPU with 96GB memory. The batch size is set to 16. The model is trained for 100000 steps. Training is performed with bfloat16 mixed precision using AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and gradient clipping of 1.0. The training time for each control condition is approximately 45 hours.

Dataset. All experiments are conducted on the proposed MiCoGen dataset. We used 1,000,000 samples for training and 19,124 samples for evaluation.

Baselines. We compare HumanFlow against representative diffusion-based baselines (ControlNet [Zhang et al. 2023], T2I-Adapter [Mou et al. 2024]) and recent flow-based controllable generation frameworks built on FLUX.1-dev [Labs 2024] (OminiControl [Tan et al. 2025], EasyControl [Zhang et al. 2025b]). Baseline models are initialized from their official pretrained checkpoints and further fine-tuned on MiCoGen.

Metrics. We evaluate model performance from three complementary aspects. (1) Image Quality. Fréchet Inception Distance (FID) [Heusel et al. 2017] and HPSv2 [Wu et al. 2023] are used to assess visual realism and perceptual quality. (2) Text-Image Alignment. CLIP score [Hessel et al. 2021] is adopted to measure semantic consistency between generated images and text prompts. (3) Control Alignment Metrics. We evaluate controllability using task-specific structural metrics. Edge adherence is measured by the F1 score on Canny maps. Pose accuracy is evaluated using COCO Object Key-point Similarity (OKS). For region-level controls, including semantic segmentation and densepose, we use mean Intersection-over-Union (mIoU). Depth is assessed via human-region mean squared error (MSE), while surface normal consistency is measured by the proportion of pixels whose angular error falls below predefined thresholds. All metrics are computed within human regions to ensure structure-aware and fair evaluation.

4.2 Quantitative evaluation

Evaluation based on Canny conditions. Canny edges impose low-level boundary constraints that emphasize contours and fine-grained structures. As shown in Table 3, diffusion-based methods struggle to faithfully follow edge guidance, resulting in high FID and inferior structural alignment. FLUX-based approaches substantially improve both visual quality and edge adherence. HumanFlow achieves the best overall performance, with the lowest FID and the highest HPSv2, CLIP, and F1 scores, indicating superior contour fidelity and semantic consistency.

Table 3. Quantitative comparison under Canny edge conditions.

Method	FID↓	HPSv2↑	CLIP Score↑	F1↑
ControlNet	68.73	0.2753	30.06	0.86
T2I-Adapter	47.49	0.2741	30.36	0.85
OminiControl	34.58	0.2842	30.49	0.93
EasyControl	34.68	0.2843	30.86	0.92
Ours	33.80	0.2889	31.47	0.95



Fig. 7. Qualitative comparison of human image generation given a canny and a prompt.

Evaluation based on Depth conditions. As reported in Table 4, diffusion-based baselines exhibit limited ability to exploit depth cues. FLUX-based methods significantly improve performance across all metrics. HumanFlow further advances perceptual quality and geometric accuracy, achieving the highest HPSv2 and CLIP scores while maintaining competitive FID and substantially reducing depth MSE, demonstrating effective integration of global geometry.

Table 4. Quantitative comparison under depth conditions.

Method	FID↓	HPSv2↑	CLIP Score↑	MSE↓
ControlNet	73.41	0.2191	27.40	1231
T2I-Adapter	54.62	0.2769	30.34	1457
OminiControl	35.85	0.2896	30.60	1154
EasyControl	35.84	0.2869	30.54	1092
Ours	35.46	0.2962	30.63	712

Evaluation based on Keypoints conditions. Keypoints guidance requires accurate modeling of articulated human body topology. As shown in Table 5, diffusion-based methods suffer severe degradation under sparse keypoints constraints. EasyControl improves pose alignment but sacrifices perceptual quality. HumanFlow consistently achieves the best performance across all metrics, indicating robust pose controllability and anatomically coherent generation.

Table 5. Quantitative comparison under keypoint conditions.

Method	FID↓	HPSv2↑	CLIP Score↑	OKS↑
ControlNet	77.55	0.2890	29.94	0.93
T2I-Adapter	75.20	0.2882	30.01	0.90
EasyControl	36.45	0.2731	30.97	0.95
Ours	35.85	0.2985	31.15	0.96

Table 6. Quantitative comparison under segmentation conditions.

Method	FID↓	HPSv2↑	CLIP Score↑	mIoU↑
ControlNet	70.28	0.2629	30.29	0.80
T2I-Adapter	61.84	0.2702	30.07	0.78
EasyControl	36.41	0.2809	30.72	0.81
Ours	34.75	0.2840	30.93	0.84

Evaluation based on Segmentation conditions. Semantic segmentation provides dense region-level supervision. As reported in Table 6, EasyControl outperforms diffusion-based methods in structural alignment. HumanFlow further improves both image quality and region correspondence, achieving the lowest FID and the highest HPSv2, CLIP, and mIoU scores, demonstrating stable and accurate generation.

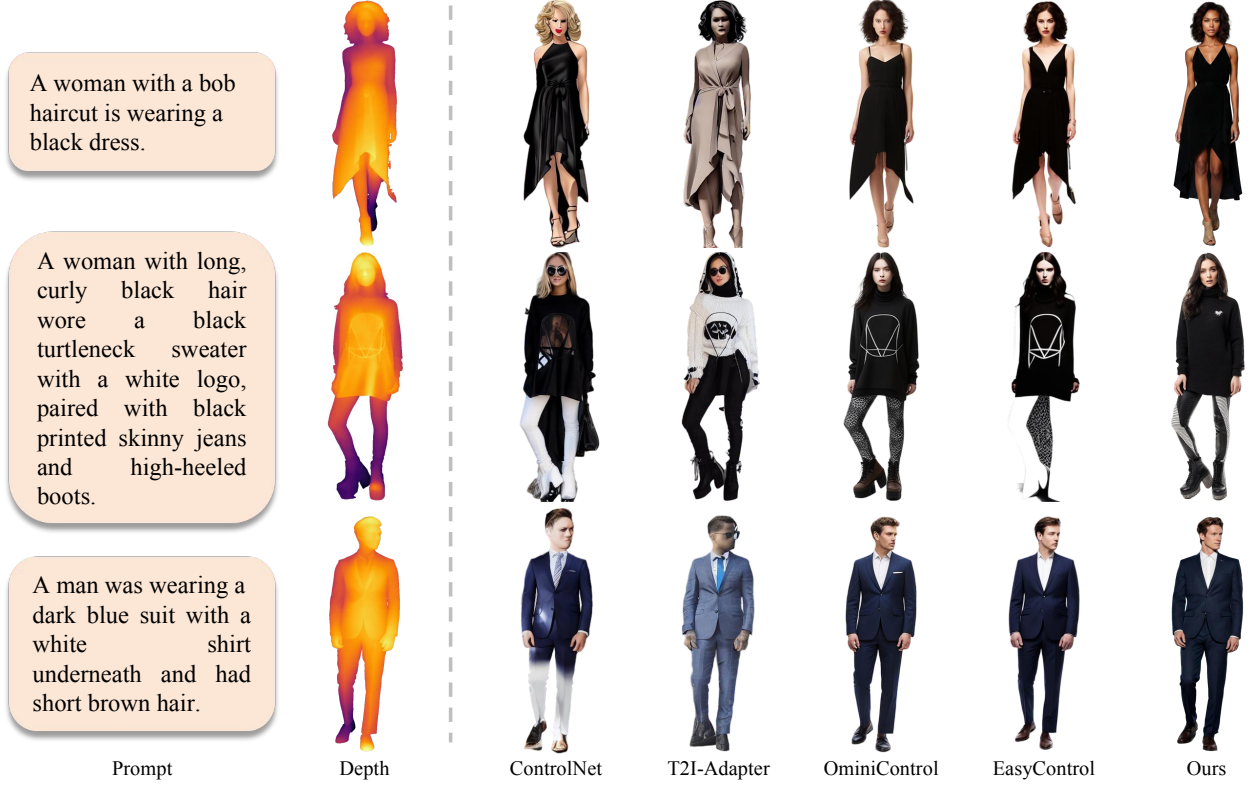


Fig. 8. Qualitative comparison of human image generation given a depth and a prompt.

Evaluation based on Normal conditions. As shown in Table 7, ControlNet exhibits severe performance degradation. HumanFlow dramatically improves all metrics, highlighting the advantage of deterministic flow dynamics for exploiting geometric control signals.

Table 7. Quantitative comparison under surface normal conditions.

Method	FID↓	HPSv2↑	CLIP Score↑	% Within t° ↑		
				11.25°	22.5°	30°
ControlNet	114.96	0.2392	27.55	75%	88%	91%
Ours	36.12	0.2903	30.95	80%	92%	97%

Evaluation based on DensePose conditions. DensePose provides dense correspondence between image pixels and the human body surface. As shown in Table 8, HumanFlow effectively handles dense human-centric structural constraints, generating coherent and anatomically consistent full-body human images.

Table 8. Quantitative results under DensePose conditions.

Method	FID↓	HPSv2↑	CLIP Score↑	mIoU↑
Ours	35.24	0.2976	30.57	0.83

Evaluation based on cross-dataset. Although our experiments are conducted on MiCoGen, HumanFlow is not tied to a specific dataset, as it conditions on generic control signals rather than dataset-specific image identities. This design enables strong generalization to unseen data distributions. Under a unified segmentation control setting, we evaluate HumanFlow on MiCoGen, DF-MM [Jiang et al. 2022], and CosmicMan-HQ 1.0 [Li et al. 2024a], as shown in Table 9 and Table 10. HumanFlow consistently outperforms prior methods across all metrics on both datasets, demonstrating improved structural consistency and visual quality in cross-dataset settings.

Table 9. Quantitative comparison under segmentation conditions on CosmicMan-HQ 1.0 [Li et al. 2024a].

Method	FID ↓	HPSv2 ↑	CLIP Score ↑	mIoU ↑
ControlNet	78.55	0.2437	28.96	0.73
T2I-Adapter	67.28	0.2501	29.15	0.74
EasyControl	39.87	0.2735	29.98	0.77
Ours	34.92	0.2816	30.59	0.80

4.3 Qualitative Evaluation

We qualitatively evaluate HumanFlow under a diverse set of human-centric control conditions, including Canny, keypoints, depth, segmentation, normal, and densepose. All results are generated using



Fig. 9. Qualitative comparison of human image generation given keypoints and a prompt.

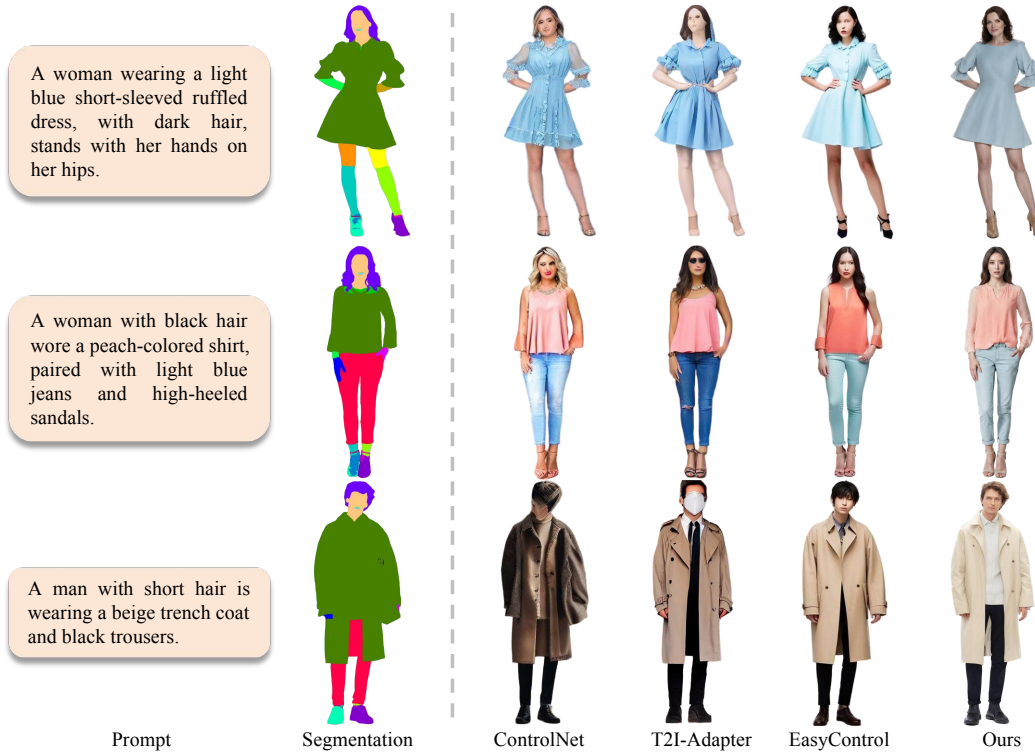


Fig. 10. Qualitative comparison of human image generation given a segmentation and a prompt.

Table 10. Quantitative comparison under segmentation conditions on DF-MM. [Jiang et al. 2022]

Method	FID ↓	HPSv2 ↑	CLIP Score ↑	mIoU ↑
ControlNet	79.42	0.2421	27.78	0.62
T2I-Adapter	73.53	0.2405	28.02	0.66
EasyControl	39.08	0.2616	29.55	0.74
Ours	35.14	0.2820	30.26	0.81

a single control modality combined with a text prompt. We highlight differences in structural fidelity, and visual coherence. More qualitative generation results can be found in the Appendix.

Canny-Guided Generation. Diffusion-based methods often exhibit broken silhouettes or blurred body outlines. HumanFlow preserves global human contours while maintaining coherent textures, indicating that deterministic flow dynamics enable stable propagation of structural constraints throughout generation.

Depth-Guided Generation. HumanFlow produces coherent body volumes and consistent depth ordering, whereas diffusion-based baselines frequently generate flattened or fragmented human structures under strong depth constraints.

Keypoint-Guided Generation. Existing methods often suffer from limb distortion or joint misalignment, particularly for uncommon poses. HumanFlow achieves higher Keypoints fidelity, preserving correct limb connectivity, proportions, and symmetry, reflecting the effectiveness of the Human Topology Consistency Loss.

Segmentation-Guided Generation. Prior methods tend to overemphasize region boundaries, leading to rigid or unnatural textures. HumanFlow maintains accurate region alignment while preserving realistic and diverse appearance.

Normal-Guided Generation. HumanFlow generates smooth and consistent shading aligned with surface orientation, while diffusion-based methods often produce artifacts or unstable results, highlighting the advantage of noise-free flow matching for geometry-sensitive conditions.

DensePose-Guided Generation. HumanFlow faithfully follows densepose inputs while maintaining globally coherent body structure and high visual quality, demonstrating robustness to dense, high-dimensional control signals.

subsectionRobustness Analysis We further evaluate the robustness of HumanFlow, as illustrated in Figure 13. These examples cover particularly difficult scenarios such as side views with limited joint visibility, back views, and side-view occlusions. Despite these challenges, HumanFlow still achieves structurally consistent alignment with the input pose while preserving fine-grained appearance details, demonstrating strong robustness to viewpoint variations and imperfect poses.

4.4 Ablation Study

We conduct ablation studies to evaluate the contributions of the Control Encoder (CE) and the proposed Human Topology Consistency

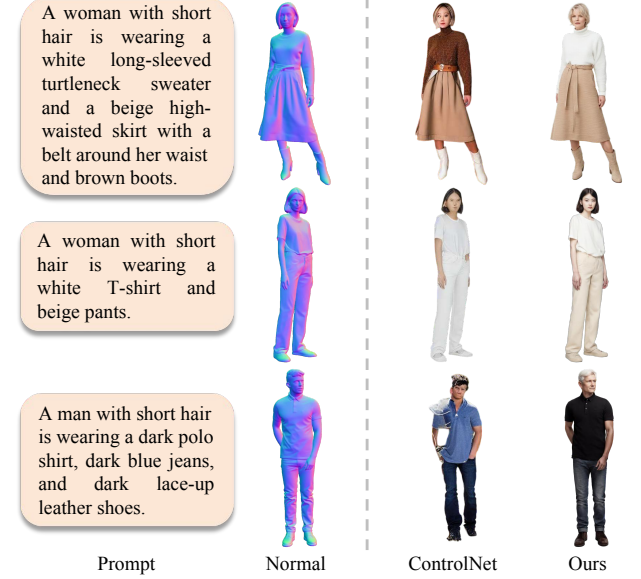


Fig. 11. Qualitative comparison of human image generation given a normal and a prompt.

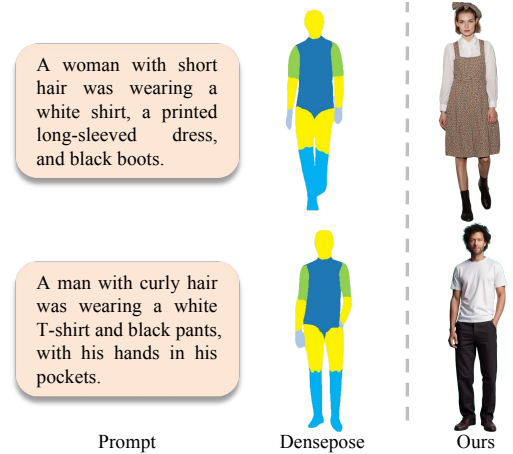


Fig. 12. Qualitative results of human image generation given a densepose and a prompt.

Loss (HTCL). Each component is removed from the full model individually. For the CE ablation, the Control Encoder is replaced with a lightweight three-layer convolutional feature extractor. All other settings, including network architecture, training data, optimization strategy, and inference configuration, are kept identical.

Effect of HTCL. As shown in Table 11, incorporating HTCL consistently improves image quality and semantic alignment across all control conditions. Compared to the variant without HTCL, the full model achieves lower FID and higher HPSv2 and CLIP scores, indicating more structurally coherent and semantically aligned generations. On average, HTCL yields a noticeable reduction in FID

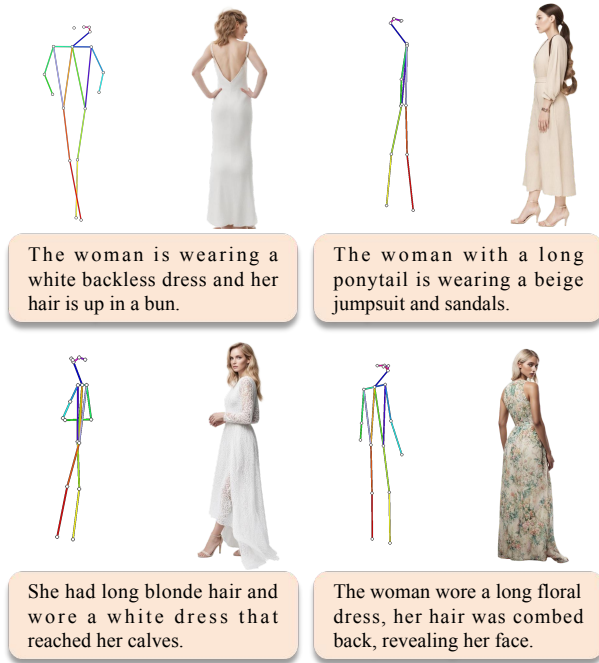


Fig. 13. Results for robustness analysis. Each example includes an input keypoint and prompt, along with the generated image. The model effectively handles challenging cases such as side views, self-occlusions, and back views, producing structurally accurate and high-quality results.

together with consistent gains in perceptual quality and text-image alignment. Qualitative results in Figure 14 further show that removing HTCL leads to structural artifacts such as implausible limb configurations, highlighting its role in enforcing anatomically valid human topology.

Effect of Control Encoder. The Control Encoder is responsible for mapping heterogeneous control signals into spatially aligned latent features for token-level injection. Removing this module results in a clear performance degradation across all control tasks (Table 11), demonstrating that a dedicated control feature extractor is critical for effective conditioning. As illustrated in Figure 15, the absence of the Control Encoder substantially weakens adherence to the specified control signals, leading to increased structural instability and misalignment between generated images and control inputs.

5 Limitations

Despite its high-fidelity and controllable full-body human image generation, HumanFlow still presents several limitations. First, MiCo-Gen relies on automatically generated control annotations. Inaccuracies in the predicted control signals may occasionally introduce localized inconsistencies in the generated results. As illustrated in Fig. 16, the normal map of missing regions leads to the generation of incorrect images. Second, while HumanFlow achieves strong structural controllability, faithfully modeling fine-grained appearance details in human images—such as clothing textures, accessories, and small-scale patterns—remains a challenging problem shared by

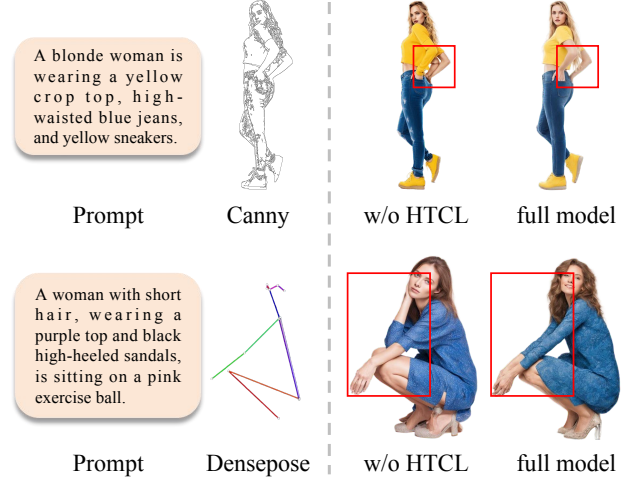


Fig. 14. Qualitative ablation of HTCL. Removing HTCL results in spurious extra limbs and distorted body topology, whereas incorporating HTCL produces improved adherence to the conditioned human layout. Red boxes highlight representative structural artifacts.

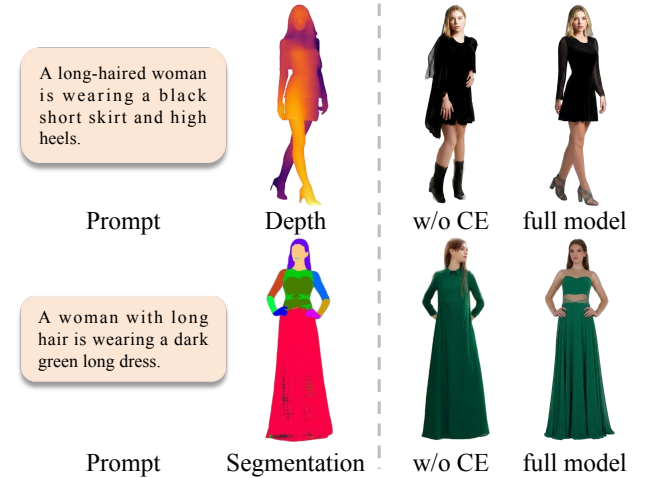


Fig. 15. Qualitative ablation of the Control Encoder (CE). Compared with the full model, removing CE under depth and segmentation guidance leads to weaker adherence to the control signals, resulting in less consistent structure. The full model better preserves both global shape and fine-grained attributes.

current generative frameworks. Finally, our study focuses on a set of widely adopted human-centric control conditions. While HumanFlow is formulated in a modality-agnostic manner, extending it to controllable generation tasks may require modality-specific conditioning designs or tailored representations to exploit the structural characteristics of different control signals fully.

Table 11. Ablation study on different control conditions, analyzing the contributions of the Control Encoder and the Human Topology Consistency Loss (HTCL). **w/o HTCL** removes the proposed topology consistency loss to evaluate its role in enforcing anatomically coherent human structures, while **w/o CE** removes the Control Encoder to assess its importance in encoding and injecting heterogeneous control signals.

Task	Method	FID ↓	HPSv2 ↑	CLIP ↑
Canny	w/o HTCL	35.10	0.2820	30.86
	w/o CE	37.25	0.2751	29.92
	HumanFlow	33.80	0.2889	31.47
Depth	w/o HTCL	35.62	0.2891	30.06
	w/o CE	37.10	0.2744	29.44
	HumanFlow	35.46	0.2962	30.63
Keypoints	w/o HTCL	36.14	0.2913	30.54
	w/o CE	37.80	0.2785	29.81
	HumanFlow	35.85	0.2985	31.15
Segmentation	w/o HTCL	35.18	0.2768	30.31
	w/o CE	39.65	0.2611	29.72
	HumanFlow	34.75	0.2840	30.93
Normal	w/o HTCL	38.46	0.2837	30.27
	w/o CE	40.20	0.2652	29.08
	HumanFlow	36.12	0.2903	30.95
DensePose	w/o HTCL	37.33	0.2841	30.29
	w/o CE	44.10	0.2724	29.15
	HumanFlow	36.12	0.2903	30.95

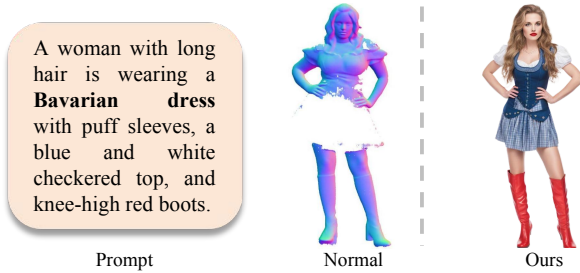


Fig. 16. Failure case. Given the prompt and the normal map, the results generated by HumanFlow fail to faithfully preserve key semantic attributes of the clothing, such as the structural characteristics of Bavarian attire.

6 Conclusion

We present HumanFlow, a flow-matching-based framework for high-fidelity and structurally controllable full-body human image generation. HumanFlow formulates generation as conditional flow matching and introduces three key components: a dedicated Control Encoder for unifying heterogeneous control signals, a lightweight Token-ControlNet for stable token-level control injection without modifying the pretrained backbone, and the Human Topology Consistency Loss for anatomically plausible human structures. Extensive

quantitative and qualitative experiments on our large-scale MiCo-Gen dataset demonstrate that HumanFlow consistently outperforms other methods in terms of visual fidelity and controllability.

HumanFlow demonstrates strong performance in controllable human image generation, but several promising directions remain for future exploration. First, we aim to enhance fine-grained appearance modeling, such as clothing textures and accessories, through appearance priors. Second, we plan to extend HumanFlow to support more abstract semantic controls, including motion and high-level behavioral constraints. Finally, we will generalize the proposed conditional flow-matching framework to other generation tasks, such as articulated animals and human-object interaction scenarios.

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